

Chapter 13

Understanding and Promoting Thinking About Knowledge: Origins, Issues, and Future Directions of Research on Epistemic Cognition

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Epistemic cognition is the thinking that people do about what and how they know. Education has long been concerned with promoting reflection on knowledge and processes of knowing, but research into epistemic cognition began really in the past half century, with a tremendous expansion in the past 20 years. This review summarizes the broad range of psychological and education research that comprises the study of epistemic cognition, and it identifies various fault lines that currently prevent coherent synthesis of theoretical models and empirical findings. The fault lines include differences in how scholars conceptualize knowledge and cognition, and the contextual nature of epistemic cognition, with consequent differences in accounts of individual development, as well as in research methods. In the coming century, research that can integrate findings among individual, situative, and cultural accounts of cognition may enable the advancement of coherent models of epistemic cognition and its development and support improved educational efforts aimed at such development.

The most prominent and influential trend in the still-young 21st century is the Digital Age's rapid production and distribution of information (Leu, Kinzer, Coiro, Castek, & Henry, 2013). Unfortunately, this trend brings with it age-old concerns about the quality, veracity, and utility of information. Citizens of the modern world must have the dispositions, beliefs, and skills to think critically about others'

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claims and the ability to make convincing claims of their own (Goldman et al., 2010; National Education Association, 2014). In a time of unprecedented information complexity, the evidence that people often do not think about that complexity in effective ways (Sinatra, Kienhues, & Hofer, 2014; Stanovich, 2010) has led to numerous calls for educators to prioritize teaching people how to discern positive contributions to human knowledge from opinion, dogma, and unsubstantiated or maliciously intended deceptions (Council of Chief State School Officers, 2010; NGSS Lead States, 2013).

The study of *epistemic cognition* concerns how individuals think about what they know, what knowledge is, how it can be used, and how they know what they know (Greene, Sandoval, & Bråten, 2016a; Kitchener, 1983). Questions about knowledge and knowing have, for most of history, been the purview of epistemology—the philosophical study of knowledge and its justifications. Over the past century, psychological inquiry into how individuals think about epistemic issues has steadily grown, as it has become apparent that epistemic cognition influences learning and reasoning. This review summarizes the broad areas of psychological and education research that have combined to define the area of research on epistemic cognition. This work has old roots in philosophical epistemology but has expanded tremendously in the past 50 years as its importance to education has been increasingly recognized. Our review identifies a number of fault lines in the various approaches to studying epistemic cognition. We characterize these fault lines and propose avenues toward their resolution that can productively guide research on epistemic cognition over the next century.

APPROACH TO REVIEW

The study of epistemic cognition is not a single, coherent field. Rather, scholars from educational and developmental psychology, disciplinary education, and the learning sciences all have been interested in how learners come to know and to think about knowledge. Reviewing the disparate breadth of relevant work presents unusual challenges. A straightforward review of the history of each field now considered relevant to epistemic cognition would be extremely lengthy, making it difficult to identify points of confluence, divergence, and incompatibility. Instead, we briefly review historical developments within these disparate fields to describe the origins of key conceptualizations and empirical findings relevant to epistemic cognition. We then shift away from this historical overview to draw out what we see as major fault lines running across the fields of epistemic cognition research. The fault lines represent obstacles to producing more integrated work on epistemic cognition—integration we believe necessary both to intellectual efforts to model epistemic cognition and to educational efforts to develop it. This approach will identify the major models of epistemic cognition applied to education, their conceptual and empirical strengths and weaknesses, and areas of scholarship that can strengthen and develop these models and their educational applications. We conclude by suggesting how the fault lines might be addressed in the coming century in ways that can support more integrative work across the many fields that concern themselves with epistemic cognition.

To gather material relevant to our review, we conducted a search within the following Thomson Reuters Web of Science databases: Social Sciences Citation Index; Conference Proceedings Citation Index—Social Sciences & Humanities; Book Citation Index—Social Sciences & Humanities. We used the search terms *personal epist**, *epistemic cognition*, and *epistem* belief**, which returned 294, 29, and 724 results, respectively. Thomson Reuters produced reference and citation data on each publication. We included in our review database all publications cited at least 10 times, resulting in 225 unique results. We expanded our results list by entering all chapters from three prominent texts on epistemic cognition: Bendixen and Feucht (2010), Hofer and Pintrich (2002), and Khine (2008). We also added Perry's (1970) seminal text on intellectual and ethical development. Finally, we included chapters from the new *Handbook of Epistemic Cognition* (Greene, Sandoval, & Bråten, 2016b). This resulted in a final database of 267 publications.

The Thomson Reuters index is a limited subset of all possible publications (e.g., it does not capture book chapters well), which may lead to misrepresentations of how the 267 publications influenced the field. Google Scholar captures a broader array of publications (e.g., book chapters, books, white papers), making it a useful comparison tool for evaluating publications' impact. Only 5 of our 267 publications were not located in Google Scholar. We calculated total citations and average citations per year and ordered rankings for each publication from both the Thomson Reuters and Google Scholar sources.

As expected, the Thomson Reuters data set was more conservative than the Google Scholar data set. Thomson Reuters indicated that our database of epistemic cognition literature averaged 44.75 citations per publication, with an *h*-index of 54. Google Scholar data resulted in 159.09 citations per publication, with an *h*-index of 101. These indices reflect the high volume of activity in epistemic cognition research. The values and ranks between data sets were highly correlated. The Pearson correlation for total citations between data sets was .966; the Pearson correlation for average citations per year was .928. Using Spearman's rho for ranked data, the correlation for total citations rankings was .903; and for the average citations per year rankings, the correlation was .834. All correlations were statistically significant at $p < .001$. Our hand review of the rankings across databases corroborated the high degree of statistical correlation between the two, as well as corroborating our own intuitions about influential publications. Therefore, when deciding on the most significant publications to include in our review, we consulted our citation rankings as well as our own sense of influential work.

ORIGINS OF RESEARCH ON EPISTEMIC COGNITION

The sections that follow provide a historical overview of various strands of scholarship on questions and concerns related to epistemic cognition, although some of the fields discussed below did not explicitly frame themselves in such terms. Given the number of fields in psychology and education addressing epistemic issues, we do not

attempt a comprehensive review of any single one of them. Instead, we highlight research findings and conceptual models that have influenced research on epistemic cognition over time. These sections rely more heavily than later ones on secondary sources (i.e., prior reviews), where interested readers can find much more detail on any particular field.

Philosophical Epistemology

In many of the most cited sources in our database, the authors drew from philosophical epistemology to define epistemic cognition and determine its scope (Chinn, Buckland, & Samarapungavan, 2011; Greene, Azevedo, & Torney-Purta, 2008; Hofer & Pintrich, 1997; Muis, Bendixen, & Haerle, 2006). In the first sentence of their landmark review of epistemic cognition research, Hofer and Pintrich (1997) defined epistemology as “an area of philosophy concerned with the nature and justification of human knowledge” (p. 88). This definition guided their conceptualization of epistemic cognition, and many others as well (e.g., Greene et al., 2008; Muis et al., 2006). Chinn et al.’s (2011) recent argument for an expansion of the scope of epistemic cognition research was based on their review of contemporary philosophical epistemology, in which they found that “epistemologists view their field much more broadly” (p. 145) than previous education researchers had suggested. Clearly, philosophical epistemology has influenced, and continues to influence, education research on epistemic cognition.

It is far beyond the scope of this chapter to comprehensively review epistemology, a field with roots stretching back to Plato.¹ Instead, we describe how ideas from “traditional” epistemology (Rysiew, 2016) have informed the scope and foci of epistemic cognition research. Then, we discuss how more recent “naturalized” epistemology scholarship has pushed researchers to consider more social and situated views of epistemic cognition.

Traditional Epistemology

Epistemology (from the Greek, “study of knowledge”) has, for most of its history, focused on the proper means of defining knowledge and discerning it from opinion, faith, and misinformation (Steup, 2014). Until the late 20th century, the predominant view of necessary and sufficient conditions for claiming knowledge was that the claim had to be a justified true belief (Chisholm, 1983): For a person to have knowledge, a proposition must be true, and the person must believe it to be true. Philosophers recognized that a person could have a true belief by chance, such as believing that this sentence was written on an Apple computer, while lacking any reasonable cause to make the claim. To avoid characterizing cases of true belief by chance or luck as knowledge, philosophers added that to know something, one had to be justified in that knowledge, for example, by seeing one of this chapter’s authors typing on an Apple computer (i.e., justification by perception).

Debates in traditional epistemology have focused largely on the proper and sufficient means of justification, with some arguing for evidence (e.g., perception) as

being sufficient (Connee & Feldman, 1985) and others arguing that justifications must come from processes that have been shown to reliably produce knowledge (e.g., testimony from a trusted source; Goldman, 1994). Gettier (1963) issued a major challenge to traditional epistemology when he showed that having a justified true belief alone was not a sufficient condition for knowledge, leading to increased focus on other means of justification or delimitation of knowledge (Moser & vander Nat, 1987). Another challenge to the justified true belief model came from radical skepticism, the idea that humans can never know anything with certainty, debated in epistemology since Plato. The details of Gettier's (1963) arguments and how they have been debated in the field are beyond the scope of this review (cf. Williams, 2001). The key point is that until fairly recently, the majority of education research on epistemic cognition, influenced by traditional epistemology, focused on when and how individuals recognize that knowledge claims are complex, evolving phenomena requiring justifications (e.g., Hofer & Pintrich, 1997; Kuhn, Cheney, & Weinstock, 2000), and which types of justifications are associated with positive academic outcomes (e.g., Bråten, Britt, Strømsø, & Rouet, 2011; Greene et al., 2008; Greene, Torney-Purta, & Azevedo, 2010).

Naturalized Epistemology

Many contemporary epistemologists continue to explore and debate traditional views of epistemology (e.g., strategic reliabilism; Bishop & Trout, 2008). Yet, partially in response to the seemingly intractable nature of Gettier problems and radical skepticism, relatively recently some philosophers have taken a different approach to epistemology (Chinn et al., 2011). One major area of scholarship is naturalized epistemology (Goldman, 1994; Kitcher, 1993; Kornblith, 1985; Quine, 1969). Adherents of this movement have argued that the field should move away from solely a priori theorizing about knowledge and knowing and, instead, bolster philosophical epistemology with findings from psychology and other sciences regarding the actual processes, cognitive and otherwise, that laypeople and experts use to make and evaluate knowledge. Rather than using abstract examples to explore their ideas (e.g., "this sentence was written on an Apple computer"), they study how disciplinary or domain ideas are justified (e.g., astronomers' decision-making processes that led them to change Pluto's status from a planet to a dwarf planet). Naturalized epistemology often intersects with social epistemology (Fuller, 1988), which is the study of how knowledge is constructed, substantiated, and communicated within and among communities. For example, communities cohere around particular epistemic practices to make and disseminate knowledge (e.g., peer review), and both philosophers and education researchers have begun studying how epistemic practices are adopted, or not, by people in those communities (Goldman, 1999; Kelly, 2014). Philosophical scholarship in naturalized epistemology, with its focus on how knowledge is determined and disseminated within and among communities of practice, coheres well with, and has informed, models of epistemic cognition based in the disciplines (e.g., Hammer & Elby, 2002) as well as those with a situated view (e.g., Chinn et al., 2011).

Philosophical Epistemology and Education

Traditional epistemology continues to influence education research on how people think about knowledge and justification (e.g., Barzilai & Weinstock, 2015; Muis & Duffy, 2013). Likewise, the naturalistic turn has changed and broadened philosophical epistemology in numerous ways, with similar effects appearing in epistemic cognition research. Questions about the development of epistemic cognition and the level of generality of means of justification have been approached, in part, by looking to studies of the knowledge work of particular disciplines (e.g., Duschl's, 2008, summary of these trends in science). Naturalized epistemology has also been used to argue for an expansion of the purview of epistemic cognition research beyond knowledge sources and justifications (Chinn et al., 2011), and to motivate study of specific forms of epistemic cognition in particular contexts, especially disciplinary learning (e.g., Kelly, 2016) and expert practice (Samarapungavan, Westby, & Bodner, 2006). The distinctions between traditional and naturalized epistemology mirror and inform fault lines between psychological and disciplinary approaches to epistemic cognition, specifically differences in emphasis between models of epistemic cognition focusing on epistemic beliefs or theories (e.g., Hofer & Pintrich, 1997) and those that focus on epistemic standards and practices in context (e.g., Chinn, Rinehart, & Buckland, 2014).

Education and Psychology

Education has always been concerned with how people come to understand the nature of knowledge and develop strategies to justify belief. Plato's *The Republic* (1991) outlined an epistemology of ideal knowledge and an educational model to obtain it; the Socratic dialogues articulated a view on how knowledge could be obtained and justified. In the modern era, Dewey (1910) expounded on the value of developing "scientific" habits of mind, and modern psychologists have routinely explored how insights from the study of knowledge generation and application are relevant in educational settings. Here we take a historical approach to highlight salient models, findings, and trends in broad strands of research related to epistemic cognition: personal epistemology research as it has been conducted within educational and developmental psychology, scholarship on epistemic aspects of cognitive development from developmental psychology, and disciplinary education research.

Personal Epistemology

The field of personal epistemology began with Perry's (1970) study of the changes in "intellectual development" of a sample of (male) Harvard students over their college careers. From this root have grown three prominent models of epistemic cognition that dominate the top 20 sources in our database, in terms of both total citations and average citations per year. Besides Perry's (1970) model, they are Kuhn et al.'s (2000) model of epistemological understanding, Schommer-Aikins's (2004) epistemological belief system, and Hofer and Pintrich's (1997) epistemological theories model. These models group into two classes, often called *developmental* (i.e., Kuhn et al., 2000;

Perry, 1970) and *dimensional* (i.e., Hofer & Pintrich, 1997; Schommer-Aikins, 2004). Developmental models characterize differences in epistemic cognition in terms of phases or stages, positing sequential movement from one stage to the next with concomitant changes to people's views of knowledge and knowing. Dimensional models, on the other hand, characterize people's views of knowledge and knowing in terms of multiple beliefs or theories, which can change independently from one another.

Perry (1970) described a scheme akin to a Piagetian stage model, with nine sequential positions grouped into four main stages, each of which included predictions about how college students would interpret all of their experiences, regardless of domain or discipline. People in the first stage of Perry's model were termed dualists, and they were posited to have an objective view of the world, believing that knowledge is factual, unchanging, and the property of experts who pass it along to others. Such views meant that dualists would not engage in critical thinking about knowledge or its sources. Through cognitive disequilibrium resulting from experiences in education and elsewhere, Perry argued that people eventually move into a multiplistic view, believing there are no objective standards for knowledge, making all views equally valid, akin to radical subjectivity. Some people, if they continue to encounter experiences fostering cognitive disequilibrium, were thought to move into the relativist category, acknowledging the contingent and contextual nature of knowledge and the necessity of using one's own logic and reasoning (i.e., critical thinking) to sort knowledge propositions from other types of claims. Perry's fourth category, commitment within relativism, largely concerns how individuals adopt particular values and identities that shape their reasoning. While Perry did little more to investigate or expand this model, his work inspired a number of other developmental models of epistemic cognition (e.g., Baxter Magolda, 1992; Belenky, Clinchy, Goldberger, & Tarule 1986; King & Kitchener, 1994).

Kuhn's (1991) work on informal reasoning about everyday problems resulted in a model that paralleled Perry's trajectory through objective and subjective views of knowledge. Absolutists view claims as facts that describe or explain a directly knowable, certain, and objective reality. These people do not see a need to think critically beyond determining which sources are expert and which knowledge claims match reality. Multiplists deny that reality can be directly knowable and view claims as subjective opinions, thus making critical thinking irrelevant. Finally, evaluativists share multiplists' view that reality is not directly knowable, but like Perry's committed relativists, they believe that critical thinking can be used to determine which claims are more justifiable than others. Kuhn's characterization of the stages of epistemic cognition development has informed numerous other publications in our database (e.g., Barzilai & Zohar, 2012). Another of Kuhn's major contributions to the field of epistemic cognition was to distinguish domains of value judgments (e.g., personal taste, aesthetics) from truth judgments (e.g., social and natural sciences) and provide evidence that developmental levels might vary across these domains.

In a major shift in conceptualization of epistemic cognition, Schommer (1990) proposed a model composed of multiple, somewhat independent belief dimensions,

on which individuals might vary. These dimensions have naïve and sophisticated poles, with the assumption that more sophisticated beliefs are associated with better performance on various learning outcomes. The names of the dimensions have changed over time, but are most commonly referred to as *simple knowledge*, *certain knowledge*, *source of knowledge*, *ability to learn*, and *quick learning* (Schommer-Aikins, 2004). The latter two belief dimensions have proved controversial, as several authors have argued that they are not epistemic in nature but, rather, concern beliefs about learning (Greene et al., 2008; Hofer & Pintrich, 1997; Sandoval, 2009). The other three beliefs roughly translate the ideas characterizing developmental models of epistemic cognition (e.g., Kuhn et al., 2000; Perry, 1970) into somewhat independent dimensions. For example, absolutists in Kuhn et al.'s model could be described as believing knowledge to be simple and certain, with the source external to the self, whereas evaluativists would view knowledge as complex and dynamic, and deriving from the self. A key difference between developmental and dimensional models is that the former posit that these beliefs change in a systematic way, whereas the latter posit that people might develop asynchronously across these dimensions.

Another of Schommer's major influences on the field was her development of a self-report instrument for measuring epistemic beliefs that produced scores amenable to quantitative analyses. Her epistemological questionnaire (EQ) ignited an order-of-magnitude increase in scholarly publications about epistemic cognition, with many of the most cited empirical studies of epistemic cognition in our database including the EQ or some variant of it (e.g., Kardash & Scholes, 1996; Schommer, 1993; Sinatra, Southerland, McConaughy, & Demastes, 2003). These publications provided evidence of correlations between the various belief dimensions of epistemic cognition and numerous desirable educational outcomes including cognitive and metacognitive strategy use (Cano, 2005), reading comprehension (Kardash & Scholes, 1996), conceptual change (Qian & Alvermann, 1995), and academic performance (Schommer, 1993; Schommer, Crouse, & Rhodes, 1992). These findings suggested that interventions designed to facilitate students' adoption of adaptive epistemic beliefs (e.g., knowledge is complex and dynamic) might have concomitant positive effects on learning and performance (Conley, Pintrich, Vekiri, & Harrison, 2004).

Aside from Perry's (1970) initial text, the most cited publication in our data set, by a large margin, is Hofer and Pintrich's (1997) review of epistemic cognition models. Hofer and Pintrich synthesized and juxtaposed the prominent developmental and belief dimension models of epistemic cognition, finding a significant amount of coherence and overlap. They used this review to warrant their epistemological theories model, composed of four somewhat independent belief dimensions: *simple knowledge*, *certain knowledge*, *source of knowledge*, and *justification of knowledge*. They classified the first two dimensions as beliefs about the nature of knowledge and the latter two as beliefs about the nature of knowing; they characterized all four as personal theories. At that time in the literature, respondents who reported relying on authorities for justification were classified as naïve, but this characterization

would be challenged in the early years of the 21st century. Hofer and Pintrich posited that change along these dimensions might be due to cognitive disequilibrium, but pointed to models of conceptual change (Posner, Strike, Hewson, & Gertzog, 1982) and sociocultural enculturation (Vygotsky, 1962) as promising analogs for epistemic development.

Dominant themes among the empirical studies of developmental and dimensional models of epistemic cognition in our database include findings that (a) epistemic cognition predicts academic performance (e.g., Cano, 2005; Schommer et al., 1992); (b) it correlates with other major constructs in education, such as implicit theories of intelligence and self-regulated learning (e.g., Bråten & Strømsø, 2005; Bromme, Pieschl, & Stahl, 2010); (c) there is questionable construct validity of scores from quantitative measures of epistemic cognition (e.g., DeBacker, Crowson, Beesley, Thoma, & Hestevold, 2008); and (d) there is a domain-specific and even task-specific aspect of epistemic cognition, according to a growing corpus of findings (e.g., Buehl, Alexander, & Murphy, 2002; Greene & Yu, 2014; Hofer, 2000, 2004; Muis et al., 2006). Influential conceptual expansions of epistemic cognition include elaborations of the relationships between epistemic cognition, metacognition, and self-regulated learning (e.g., Barzilai & Zohar, 2014; Greene, Muis, & Pieschl, 2010; Hofer, 2004); expanded models of epistemic cognition development (e.g., Bendixen & Rule, 2004); studies of how people evaluate sources of knowledge (e.g., Braasch, Bråten, Strømsø, Anmarkrud, & Ferguson, 2013; Porsch & Bromme, 2011); investigations of teachers' epistemic cognition (Bråten & Ferguson, 2015; Buehl & Fives, 2009, 2016; Fives & Buehl, 2008; Gregoire, 2003); and the expansion of Hofer and Pintrich's (1997) justification dimension to account for multiple means of warranting knowledge claims. This expansion has included the recognition that justification by authority is not necessarily naïve, and in fact is often availing and even necessary given the proliferation of knowledge claims in the modern world (Bromme, Kienhues, & Stahl, 2008; Chinn et al., 2011; Greene et al., 2008; Kitcher, 1993). Bråten et al. (2011) have demonstrated how models of multiple document comprehension can inform epistemic cognition research, particularly in terms of how people coordinate multiple means of justification.

Finally, Chinn and colleagues (Chinn et al., 2011; Chinn et al., 2014; Chinn & Rinehart, 2016) have argued that contemporary work in philosophical epistemology has a number of implications for models of epistemic cognition, including that (a) epistemic cognition is fundamentally a social phenomenon; (b) investigations of people's epistemic aims for a task are needed to contextualize and understand their actual epistemic cognition; (c) epistemic practices or reliable processes (e.g., experimental methods, sourcing) are key aspects of epistemic cognition; and (d) epistemic cognition is situated in context. Chinn and colleagues' incorporation of ideas from naturalized epistemology has challenged the traditional epistemology foundations of developmental and dimensional models. At the same time, the naturalistic turn has highlighted potentially generative connections between personal epistemology research and work in developmental psychology and disciplinary education.

Developmental Foundations of Epistemic Cognition

For many years after Perry's (1970) seminal work, epistemic cognition was presumed to develop only in adolescence or later. Burr and Hofer (2002), among others, have found evidence of epistemic cognition in young children, suggesting an early start to its development. Montgomery (1992) argued that as children pass cognitive developmental milestones in early childhood, they can be said to develop a "folk epistemology," an intuitive, perhaps naïve, way of thinking about what and how they and others know. Over the past three decades, developmental psychology research in several areas has expanded the view of antecedent foundations of epistemic cognition. Children develop and apply a range of strategies to evaluate their own and others' claims to knowledge, even though they may not understand them in explicitly epistemic terms. Our review suggests four major areas of development where connections to epistemic cognition have been made most clearly.

Theory of mind. Children's theory of mind involves the understanding that people can have mental states (e.g., beliefs, desires, goals, knowledge) that differ from others' mental states and differ also from reality (Wellman, 2011). Early research suggested that theory of mind did not develop until age 3 or later, but more recent evidence indicates even infants are able to display integral aspects of theory of mind (e.g., understanding other peoples' goals and perceptions and how they drive their behavior; Sodian & Kristen, 2016). These findings can illuminate investigations of young children's epistemic cognition, where researchers have found that 3-year-olds can exhibit evidence of a preabsolutist stage, called egocentric subjectivity. In this stage, children believe their perspective is the only one possible (Burr & Hofer, 2002; Wildenger, Hofer, & Burr, 2010). A key milestone in theory of mind development, and likely epistemic cognition as well, is children's understanding that other people may believe something to be true that the child knows is false. This understanding of false belief manifests around the fifth year of life, on average, and may serve as foundation for the kinds of source evaluations posited in many models of epistemic cognition. At a minimum, theory of mind and epistemic cognition share some common developmental foundations (e.g., attribution of mental states to others), suggesting the need for more research connecting these two areas of study (Hofer & Bendixen, 2012; Sodian & Kristen, 2016).

Causal reasoning. Children display an early, even immediate, ability to draw causal inferences about events they perceive and experience (e.g., what makes a marble roll in a straight or diagonal path). Inference develops into more systematic reasoning throughout childhood (Moshman & Tarricone, 2016). By 8 or 9 years of age, children distinguish inference from other sources of knowing, like guessing or direct perception (Pillow, Hill, Boyce, & Stein, 2000). Beyond early childhood, children also distinguish causal explanation from logical necessities or proofs (Moshman & Tarricone, 2016), and people's evaluations of causal explanations seem to shift over time from a reliance on causal mechanisms alone toward mechanisms with evidence

(Kuhn, 2001). Causal reasoning is clearly an antecedent to epistemic cognition. It may even be where epistemic cognition begins, as young children begin making justifications using perception and logic to decide, even if tacitly, which causal inferences are reliable and which are not.

Scientific thinking. The developmental study of “scientific” thinking began in the 1980s, following work with adults by Wason (1960) and others (Zimmerman, 2000). Historically, developmental psychology has framed scientific thinking in terms of hypothesis testing, experimental control of variables, and reasoning about correlation (a narrower notion of scientific thinking than is pursued in science education, see below). The biases displayed by adults to distort evidence and confirm already-held beliefs appear early in childhood and have been labeled as an inability to distinguish claims from evidence (Kuhn, Amsel, & O’Loughlin, 1988). Research suggests that a basic distinction between claims and the evidence that might bear on them, as distinct epistemic entities, emerges by school age (Ruffman, Perner, Olson, & Doherty, 1993; Sodian, Zaitchik, & Carey, 1991), but children’s competence at evaluating specific claims or the quality of evidence for a specific claim depends on familiarity with the domain and other factors (Lehrer & Schauble, 2006; Sandoval, Sodian, Koerber, & Wong, 2014). Scientists themselves display the same sorts of biases related to evidence as do lay adults (Chinn & Brewer, 1993), suggesting that what develops is not a general sense of the nature of evidence or of its role in evaluating knowledge but competence within particular domains or topics. The crucial epistemic questions therefore seem to be how individuals and groups (e.g., disciplines) develop processes of justification, source evaluation, and so forth, as well as how those processes are communicated and refined.

Social cognition. Developmental psychologists have long been concerned with how children come to trust or mistrust particular sources of knowledge (for reviews, see Clément, 2016; Harris, 2007, 2012). Children are largely dependent on other people’s testimony for gaining knowledge about the world (Fricker, 1995), hence the social nature of children’s knowledge acquisition processes (Clément, 2016). Children’s trust in others’ testimony is not blind, however. It is based on basic evaluations of trustworthiness referring to features such as the accuracy, competence, coherence, audience reception, and benevolence of sources (Clément, 2016; Harris, 2012). For example, when two sources make conflicting claims about the name of an unfamiliar object, 3- and 4-year-olds have been found to display selective trust based on the accuracy of those sources in naming a familiar object in the past (Corriveau & Harris, 2009; Jaswal & Neely, 2006; Koenig, Clément, & Harris, 2004; Koenig & Harris, 2005). These early evaluations of testimony and source reliability likely form the foundation for evaluating trustworthiness of sources later in life. The importance of testimony as an essential, social source of knowledge has recently been emphasized in the field of epistemic cognition (Chinn et al., 2011), and critical source evaluation leading to selective trust in sources of knowledge has been

found to facilitate students' understanding of multiple conflicting perspectives on controversial socioscientific issues (Braasch et al., 2013; Bråten, Strømsø, & Britt, 2009; Wiley et al., 2009).

Summary of Developmental Foundations of Epistemic Cognition

These areas of cognitive development appear to form the building blocks for epistemic cognition and highlight early competencies of children's thinking about knowledge and knowing that should be more thoroughly integrated into current models of epistemic cognition and research about them. Over the coming century, it should be quite fruitful for scholars across fields studying epistemic cognition to look at these, and perhaps other, aspects of cognitive development as essential facets of epistemic cognition. While some scholars of development (e.g., Kuhn, 2001) explicitly locate their research across psychology and education, a broader exchange of ideas across disciplines could inform conceptualizations of epistemic cognition and its development across both schooled and unschooled settings.

Disciplinary Education

Education in the disciplines of math, science, and history has been concerned with the epistemological aspects of these disciplines for decades. Historically, educational psychology research in school subjects has focused on epistemic beliefs and their relation to learning processes and academic outcomes (e.g., Depaepe, De Corte, & Verschaffel, 2016; Qian & Alvermann, 1995; Schommer et al., 1992). Researchers in the learning sciences have focused instead on relations between engagement in disciplinary practices (e.g., scientific inquiry, historical argument) and understanding of disciplinary knowledge and disciplinary epistemologies (e.g., Rosebery, Warren, & Conant, 1992; Schoenfeld, 1992; Wineburg, 1991). There are a number of recent reviews of research related to epistemic cognition in each of these disciplines, including the long-standing research on conceptions of the nature of science (Elby, Macrander, & Hammer, 2016; Lederman, 2007), and on epistemic cognition in mathematics (Depaepe et al., 2016; Muis, 2004) and history (VanSledright & Maggioni, 2016). Here we summarize trends from these disciplinary lines of research as they apply to the broader study of epistemic cognition.

One feature of epistemic cognition research in the disciplines is that it has been framed within discipline-specific epistemological concerns. Sometimes these overlap with the more general conceptions from personal epistemology research; for example, the concern in science education that students understand the tentativeness of science knowledge maps onto the general dimension of certainty of knowledge. Yet, within disciplinary education, the epistemologies of the disciplines are fleshed out more deeply and specifically than is typically the case in the psychology or personal epistemology literatures. Science educators link tentativeness to the theory-laden nature of observation in science (Lederman, 2007), while asserting that some claims are more certain than others (Osborne, Collins, Ratcliffe, Millar, & Duschl, 2003).

Historians locate uncertainty in the distance between the present and the past and the ambiguity of historical objects (VanSledright & Maggioni, 2016). In mathematics, the issue of certainty presents as whether one believes mathematical procedures are given or constructed (Depaepe et al., 2016).

The later half of the 20th century saw in all these academic disciplines a growing recognition that students need to learn the epistemology of a discipline in addition to its theories, concepts, and facts (Duschl, 2008; Schoenfeld, 1992; Wineburg, 1991). This concern has driven a range of efforts to assess students' understanding of specific epistemological conceptions within the disciplines, as well as to examine how constructs from personal epistemology relate to learning in the disciplines. At the same time, researchers have become increasingly interested in characterizing how conceptions of knowledge and knowing in the disciplines are related to particular sense-making practices in those disciplines. A range of studies have shed light on beliefs and thinking about knowledge and knowing as situated within disciplinary activities and, thus, peculiar to particular disciplines (e.g., Cobb, Stephan, McClain, & Gravemeijer, 2001; De Corte, Op't Eynde, Depaepe, & Verschaffel, 2010; diSessa, Elby, & Hammer, 2002; Greene & Yu, 2014; Hammer, 1994; Maggioni, VanSledright, & Alexander, 2009; Samarapungavan et al., 2006; Sandoval & Çam, 2011; Wineburg, 1991).

With respect to broader debates about the generality or specificity of epistemic cognition, research in the disciplines has made a strong case for disciplinary specificity tied directly to the obviously different epistemological perspectives among the disciplines. Empirical research on learning across disciplines is starting to show that students appreciate, for example, that evidence in science and history looks different (Herrenkohl & Cornelius, 2013). Likewise, across disciplines, elementary and high school students ask different types of questions when seeking to understand texts (Portnoy & Rabinowitz, 2014). For epistemic cognition research broadly, it is crucial to recognize that disciplines have developed their own epistemologies in response to the particular questions and problems of knowledge development they have faced. The likelihood of epistemic cognition predicting academic performance in these fields depends on the degree to which individuals can adopt and use discipline-specific epistemic norms and practices (Goldman, 2011; Greene, 2016).

The State of Epistemic Cognition Research

The past half-century has seen research on epistemic cognition grow immensely in a number of fields. The disparate roots of research into how people generate knowledge, interpret knowledge, and think about the nature of knowledge have begun, in the past 10 to 20 years, to converge in some ways. Current models are explicitly informed by relevant work in the field of philosophical epistemology (Chinn et al., 2011; Greene et al., 2008; Muis et al., 2006). Contemporary models of epistemic cognition also pay more attention to the antecedents and early forms of epistemic cognition suggested by work in developmental psychology (Burr & Hofer, 2002; Hofer & Pintrich, 1997). Earlier debates about the specificity or generality of

epistemic beliefs have mostly given way to a recognition that epistemic cognition has contextual and disciplinary aspects that are only somewhat understood (Sandoval, 2012), creating a need to better explain both the extent and mechanisms of generalization of epistemic cognition. The historical trends in these fields are converging to create a broader conceptualization of epistemic cognition, albeit not without some growing pains (Chinn et al., 2011; Hofer & Bendixen, 2012; Sandoval, 2012). Conceptual expansion of the idea of epistemic cognition has brought with it questions about the appropriate boundaries of the concept (e.g., Alexander, 2016). We believe these questions are important. To answer them, researchers must begin to identify and wrestle with the differences in how they conceptualize, measure, and intervene in people's engagement with knowledge and knowing.

FAULT LINES

Given the variety of intellectual perspectives on phenomena of epistemic cognition, it is no surprise that as various communities of scholarship have come into closer contact around them, a range of fault lines have emerged. These characterize dilemmas that scholars from a variety of fields have recognized as needing some resolution to move toward coherent accounts of epistemic cognition and its possible development. We highlight some fault lines that derive from our analysis of the literature we have reviewed, acknowledging that others working in this area might see different issues.

Conceptualizations of Knowledge and Cognition

One of the major intellectual shifts in the social sciences over the past decades of the 20th century and into the 21st has been the cultural turn (Bruner, 1996). In psychology, education, and philosophy, there has been a shift from viewing thinking and knowing as fundamentally individual activities to viewing them as fundamentally social (e.g., naturalized epistemology; Goldman, 1994; Quine, 1969). This shift, however, has been taken up to different extents, and differences in where scholars, and even fields, see the locus of explanation of epistemic cognition on this individual–social continuum creates a tangle of empirical results and interpretations that have yet to be clarified.

Piaget's (1972) genetic epistemology and stage theory of development directly influenced epistemic cognition research from the start. This view of knowledge construction as an individual achievement continues to underlie much of the psychology research pertinent to epistemic cognition. The term *personal epistemology* as a moniker for this research itself conveys the assumption that persons construct epistemologies as part of their efforts to interpret the world around them. The constructivist abandons the idea that knowledge is an object in the world. Instead, "*knowledge* refers to conceptual structures that epistemic agents, given the range of present experience within their tradition of thought and language, consider *viable*" (Glaserfeld, 1989, p. 124). That is, while cultural traditions of thought and language may shape experience, individual epistemic agents build conceptual structures that can viably be

considered knowledge. This is not to say that constructivists necessarily abandon the idea of an objective reality (see Phillips, 1995, for a discussion of variants of constructivism on this point), only that our knowledge of reality is constructed. The cognitive constructivism of Piaget locates that knowledge as a construct of individual minds.

Several models of epistemic cognition maintain this cognitive constructivism as the underlying definition of knowledge. Piagetian (e.g., King & Kitchener, 1994; Perry, 1970) and neo-Piagetian (e.g., Kuhn et al., 2000) models propose progressive trajectories that vary mainly in the generality they posit for the stages along the way. Belief systems models (e.g., Hofer, 2004; Schommer-Aikins, 2004) articulate dimensions of belief and their possible influences and interactions. All of these models share an assumption that cognitive representations (i.e., knowledge and belief) are constructed by individuals, via some level of reflection on experience, and then applied to subsequent experiences.

Sociocultural theories of cognition and development derived from Vygotsky (1962, 1978) locate the individual knower within a specific cultural and historical context, thereby framing knowledge as a cultural product and knowing as a cultural activity. Models of epistemic cognition that have been influenced by sociocultural perspectives vary in where they locate knowledge and knowing. What we can call a cognitive contextualist view (see especially the next section), attempts to account for how social factors influence individual knowledge construction, including individuals' ideas about knowledge and knowing. Bendixen and colleagues (Bendixen & Rule, 2004; Muis et al., 2006), for example, have proposed models that attempt to account for how environmental (i.e., social) factors and features of specific domains influence individual epistemic cognition. Muis et al. (2006) have proposed that general epistemic beliefs develop within the social and cultural context in which people grow up. Hammer and Elby (2002) have suggested that participation in culturally shaped patterns of activity lead to the construction of epistemic "resources," fine-grained cognitive structures tied closely to contexts of use. These contexts of use become bound into epistemological "frames" that organize the conditional activation of epistemic resources (Elby & Hammer, 2010). That is, in contrast to Muis et al. (2006), the epistemic resources model suggests that generalization is an outcome of repeated use.

A stronger social perspective on knowledge and knowing derives from situative perspectives on cognition (Engeström, 1987; Lave & Wenger, 1991), allied with the social epistemologists discussed above (Fuller, 1988; Goldman, 2011), who see knowledge as fundamentally a social construct and epistemology as a social phenomenon. The situative perspective takes as its base unit of analysis the individual-in-interaction, emphasizing features of interaction external to individuals and their minds. This interactional view on cognition sees knowledge as distributed among social and material resources and deployed within cultural activity. Thus, individuals' and groups' interpretations of the purpose of an activity are seen as shaping the epistemic cognition that might occur. Chinn and colleagues' (Chinn et al., 2011; Chinn et al., 2014) model thus highlights processes of epistemic cognition as tied to the epistemic aims an individual might pursue during activity, how those aims relate to

nonepistemic aims, and how aims are related to reliable processes of knowledge development. These reliable processes are social in nature: They are social practices developed within particular communities to solve specific epistemic problems (Chinn & Rinehart, 2016). The situative perspective on epistemic cognition thus sees knowledge as reflected in practice—in how people accomplish activity—rather than as a mental representation of some aspect of reality that is then applied (Kelly, 2016).

In sum, the divergent perspectives on knowledge and cognition generally are reflected in current models of epistemic cognition, not surprisingly. The crucial difference among these perspectives with respect to epistemic cognition research is that they theorize knowledge itself, its development, and related concepts such as belief and the role of context, rather differently. Such differences have implications for the unit of analysis in epistemic cognition research (e.g., individuals' construction of knowledge propositions vs. the interplay of cultural forms of knowledge and individuals' practice) as well as for perspectives on how epistemic cognition develops (e.g., a focus on changes in individuals' conceptions of knowledge vs. changes in a community's norms and practices and individual participation in those). The parallels between personal epistemology and disciplinary education models of epistemic cognition, and traditional and naturalized epistemology, are clear.

Whether such differences are incommensurate is perhaps an ideological question. Packer and Goicoechea (2000) argued that cognitive constructivism and the situative perspective are ontologically incommensurate, whereas Greeno (2015) suggested they address different levels of explanation. It is clear, however, that broad differences in how knowledge and knowing are theorized produce differences in how issues of contextualization and generality are addressed in research on epistemic cognition. They lead to different explanations of the nature of developmental pathways for epistemic cognition. They also, naturally, produce differences in research methods for identifying, defining, and studying phenomena of epistemic cognition.

Generality and Context in Epistemic Cognition

Hofer and Pintrich (1997) noted that the developmental and belief system models they critiqued paid only marginal attention to the issue of domain specificity versus domain generality in personal epistemology, with an underlying assumption that beliefs and thoughts about knowledge and knowing were largely consistent across domains and disciplines. Hofer and Pintrich (1997) discussed the possibility that both domain-specific and domain-general beliefs about knowledge and knowing existed, while suggesting there might be specific ways of thinking about knowledge and knowing within academic disciplines that made general models of personal epistemology less applicable. The idea that epistemic cognition occurs at different levels of specificity, as well as the idea that there are discipline-specific forms of epistemic cognition, has gained empirical support and been reflected in models of personal epistemology and epistemic cognition. Articles on domain or disciplinary differences in epistemic cognition were frequent among the most cited sources in our data set (e.g., Buehl & Alexander, 2001; Hofer, 2000; Kuhn et al., 2000; Muis et al., 2006).

To study generality and context, researchers working within the belief system paradigm of personal epistemology conducted between-subjects investigations (e.g., Jehng, Johnson, & Anderson, 1993; Lonka & Lindblom-Ylänne, 1996), where students majoring in different domains were compared with respect to their epistemic beliefs. They also conducted within-subject investigations (e.g., Buehl et al., 2002; Hofer, 2000), where the same students were asked about their epistemic beliefs in different domains. In general, this work supported the idea that epistemic beliefs varied as a function of academic domain and, moreover, that such variation was related to perceptions of domain structuredness. That is, researchers found that students viewed knowledge as more certain and integrated, and that they more readily accepted experts as sources of knowledge, in well-structured domains, such as mathematics, than in ill-structured domains, such as history. At the same time, however, this line of research seemed to indicate that students' epistemic beliefs were not solely domain-specific, meaning that students could simultaneously hold both domain-specific and more domain-general or overarching beliefs about knowledge and knowing (Buehl & Alexander, 2001; Muis et al., 2006).

Recent research has started to target epistemic beliefs at a topic-specific level, that is, as beliefs about more delimited subject areas (e.g., World War II) that can be subsumed under an academic domain (e.g., history). This research has led to the use of topic-specific questionnaires (Bråten, Gil, Strømsø, & Vidal-Abarca, 2009; Stahl & Bromme, 2007; Trautwein & Lüdtke, 2007) or think-aloud methodology (Ferguson, Bråten, & Strømsø, 2012; Hofer, 2004; Mason, Ariasi, & Boldrin, 2011; Mason & Boldrin, 2008; Mason, Boldrin, & Ariasi, 2010) to assess students' beliefs about specific topics along different belief dimensions (e.g., the certainty and simplicity of knowledge). For example, such topic-specific epistemic beliefs have been shown to uniquely predict comprehension when students read texts about the same topics (Bråten & Strømsø, 2010; Bråten, Strømsø, & Samuelstuen, 2008; Strømsø, Bråten, & Samuelstuen, 2008).

In the past decade, the empirical work in personal epistemology summarized above has been reflected in theoretical frameworks that take the multilevel nature of epistemic beliefs into account. For example, in their theory of integrated domains in epistemology framework, Muis et al. (2006) posited that beliefs concerning the certainty, simplicity, and source of knowledge, as well as the justification for knowing, may exist at the level of general epistemic beliefs that develop in nonacademic contexts, domain-general academic beliefs that develop in educational contexts, and domain-specific academic beliefs that reflect the various domains of knowledge that students experience. Muis, Trevors, Duffy, Ranellucci, and Foy (2016) have acknowledged that an additional level of specificity may be required beyond the domain-specific level, such as topic-specific beliefs about knowledge and knowing. In a more parsimonious model, at least with respect to levels of specificity, Greene et al. (2008) proposed that beliefs regarding the simplicity and certainty of knowledge and the justification for knowing vary at a level between domain specificity and domain generality, mainly differentiated between well-structured (e.g., mathematics and physics)

and ill-structured (e.g., history and literature) domains. In more recent work, Greene and Yu (2014) used qualitative interview data from novices and experts to critique prior conceptualizations of domain specificity in personal epistemology research; they called for greater attention to differences in knowledge and knowing across disciplines.

Although research has indicated that students' beliefs about knowledge and knowing are contextualized, in the sense that they differ across domains and topics within domains, this work is limited by the fact that it has essentially used the same questionnaires targeting the same belief dimensions across domains and topics. As a result, mean differences between scores on those dimensions across domains have been used to indicate domain specificity in epistemic beliefs, and correlations between scores on those dimensions across domains have been used to indicate domain generality in such beliefs (e.g., Buehl et al., 2002; Hofer, 2000; Muis et al., 2016). Also, most research using think-aloud methodology to assess a topic-specific level of epistemic cognition (e.g., Ferguson et al., 2012; Mason et al., 2011) has coded the resulting verbal protocols in terms of dimensions from epistemic belief system models, such as the Hofer and Pintrich (1997) multidimensional framework. Given that those measures presumed, *a priori*, the existence of a common set of dimensions or beliefs across domains, it could be argued that what has been captured within such research is not really domain- or discipline-specific epistemic beliefs and thinking but, rather, the extent to which scores on general belief dimensions vary across domains and disciplines (Elby et al., 2016; Hofer, 2006). To better understand patterns of epistemic beliefs and thinking within different disciplines, other approaches may be needed.

As indicated above, perspectives from disciplinary education suggest contextual differences both between and within disciplines. Researchers representing such perspectives argue that students' epistemic beliefs and thinking should be understood as situated within individual and collective disciplinary learning activities (Sandoval, 2012, 2015). Some research suggests that variations in epistemic aims and virtues may also be discipline specific or even situation specific. In an intriguing study in history, Gottlieb and Wineburg (2012) showed that historians' epistemic cognition, specifically the criteria they used for evaluating claims and evidence, depended on the relationships between the documentary material under consideration and the values and commitments of the historians. The same historians could switch their thinking between academically and religiously based forms of epistemic cognition, even within the same text. It seems highly unlikely that such disciplinary and situational variation can be captured by a small set of generally applicable dimensions, an assumption on which much research on the specificity-versus-generality issue within personal epistemology seems to rest (Greene & Yu, 2014).

Arguably, the theoretical framework best aligned with the discipline-specific, situated view discussed above is the contextualist resource framework of Hammer and Elby (2002; Elby & Hammer, 2010), which focuses on variations in epistemic cognition in response to situational demands. This means that learners are conceived of as

responding adaptively or maladaptively to specific tasks by activating and deactivating existing fine-grained epistemic resources. According to Elby and Hammer (2010), the activation of fine-grained epistemic resources in a specific context over time may form a stable network with belief-like qualities, an “epistemological frame” that is typically activated in that context and sometimes even generalized across contexts.

Likewise, Chinn and colleagues (Chinn et al., 2011; Chinn et al., 2014; Chinn & Rinehart, 2016) have argued for a fine-grained, context-specific approach to epistemic cognition, where epistemic cognition varies across situations and may predict situation-specific patterns of learning. At the same time, Chinn et al.’s framework is a multicomponential model that expands on previous multidimensional personal epistemology models, with a fixed set of epistemic cognition components seen as applicable across disciplines and situations. How epistemic cognition plays out within those components is viewed as largely dependent on contextual factors, however, allowing for essential variability across and within specific disciplines.

Finally, Barzilai and Zohar’s (2014) multifaceted framework, distinguishing between cognitive and metacognitive aspects of epistemic thinking, can be said to reflect recent developments toward further contextualization and situativity in epistemic cognition research. In particular, they proposed that the cognitive aspect of epistemic thinking, concerning specific information, knowledge claims, sources, and epistemic strategies, may be highly specific in the sense that it varies with task contexts. The metacognitive aspect of epistemic thinking, however, is viewed as having more pronounced domain-general qualities within this framework. Epistemic metacognition involves, among other things, knowledge about what makes a source reliable, but epistemic cognition involves using that knowledge to evaluate a particular source (see also Barzilai & Zohar, 2012).

In sum, a disciplinary view of epistemic cognition implies that general models of personal epistemology, as well as models that operate with a fixed set of dimensions across domains and disciplines, do not sufficiently capture important differences in the epistemic norms and practices across the disciplines. Such critiques imply the need for potentially extensive model revision, which can be interpreted as another fault line between the fields. To build theory bottom-up from such research, a first step could be to produce more local models applicable to a limited range of learners and contexts. In turn, such models might be combined to produce more overarching frameworks for epistemic cognition that are better grounded in actual knowledge construction and evaluation within the disciplines (Elby et al., 2016). Such models may then be used to identify commonalities or convergences in epistemic cognition across contexts.

Developmental Pathways

Despite the consistency of proposed developmental trajectories in models of epistemic cognition described earlier, evidence for a clear developmental pathway has been inconsistent, and multiple lines of evidence undermine the notion of omnibus stagelike epistemic worldviews (Hofer & Bendixen, 2012; Sandoval, 2012). This has

led to proposals that people develop domain-specific epistemic theories (Hofer, 2006) that follow the general trend identified by Kuhn et al. (2000). This line of theorizing argues that such domain-specific theories develop at variable rates, dependent on personal experiences. Within domains, however, scholars such as Hofer, Kuhn, and others (e.g., Hallett, Chandler, & Krettenauer, 2002) have argued for recapitulation of the same developmental trend from absolutist to evaluativist.

A problem for developmental models is that “evidence for just about every conceivable sort of epistemic development has been shown to characterize persons of just about every conceivable age” (Chandler, Hallett, & Sokol, 2002, p. 161). This weight of competing evidence and interpretations about epistemic development led Chandler et al. (2002) to argue that epistemic development is recursive, that people pass through epistemic levels more than once within and across domains of epistemic cognition. At the moment, it is not clear what evidence can determine the viability of the developmental recursion hypothesis. Elby et al. (2016) argued that the recursion proposal is similar to the proposal of planetary “epicycles” as an effort to rescue Ptolemaic theories of celestial motion. They suggested that their epistemic resources model provides a more parsimonious account, arguing that development is nothing more or less than the construction and coordination of epistemic resources through experience in the contexts of activity in which their use is productive.

At issue is that what appears to be a broad developmental trend may be an artifact of researchers’ own assumptions and efforts to document that trend. There is a need to account for a broad range of evidence that is at least potentially inconsistent with the reigning developmental hypothesis. Notions of epistemic development from Perry (1970) forward have been rooted in a distinction that sees development as a slow, broad, maturational process and learning as a more rapid, specific, enduring adaptation to environmental influences. This distinction is problematic for epistemic development for several reasons. First, epistemic change, whether characterized as development or not, may depend on systematic, organized schooling, which can hardly be said to be a natural maturational process. Second, describing epistemic change as independent of maturation seems to overlook the possibility that interventions to change epistemic cognition may be more or less successful depending on more general developing patterns of epistemic thinking in individuals. Finally, specific changes in epistemic cognition from particular interventions may generalize to influence epistemic developmental trajectories as described by Perry (1970) and other developmentalists (e.g., Baxter Magolda, 1992; King & Kitchener, 1994; Kuhn et al., 2000).

Much more conceptual and empirical work is needed to distinguish models and mechanisms of epistemic change from epistemic development, to the extent that scholars continue to see them as separate, albeit related, processes. Mechanisms of change have been proposed, from Hammer and Elby’s model of activation of epistemic resources and their stabilization into epistemological frames from repeated activation in context (Elby & Hammer, 2010), to Bendixen and colleagues’ constructs of epistemic doubt, volition, and resolution strategies (Bendixen & Rule, 2004; Muis

et al., 2006). More work is needed to clarify proposed mechanisms of change, including how components of such mechanisms are involved in specific instances of epistemic change, as well as the ways in which changes in epistemic cognition may generalize. Research is needed that looks at epistemic cognition across multiple contexts over longer periods of time, to connect microgenetic processes to ontogenetic ones (Sandoval, 2014).

Research Design and Methods

Research methods and designs for capturing epistemic cognition vary a great deal across the fields engaged in this work, but can be roughly differentiated based on the assessment methods used, the contexts in which the work is done, and the kinds of epistemic cognition that are targeted. The most controversial type of assessment methods used are those that rely on self-report. The majority of self-report instruments require participants to rate their degree of agreement or disagreement with various statements by using Likert-type response scales. The first such instrument was Schommer's (1990) EQ, which was followed by various iterations and variations of that instrument (e.g., Epistemic Belief Inventory; Schraw, Bendixen, & Dunkle, 2002), as well as instruments that require participants to make semantic differential judgments, such as Connotative Aspects of Epistemological Beliefs (CAEB; Stahl & Bromme, 2007). These instruments, most commonly used by adherents of dimensional models of epistemic cognition (e.g., Hofer & Pintrich, 1997; Schommer-Aikins, 2004), have been criticized for their failure to replicate posited relations between items and proposed factors (Clarebout, Elen, Luyten, & Bamps, 2001; DeBacker et al., 2008) and for their reliance on response formats (i.e., Likert-type response scales) better suited for the measurement of attitudes than beliefs or practices (Carifio & Perla, 2007; Hofer & Sinatra, 2010).

Aside from psychometric concerns, self-report assessment methods have been criticized as inauthentic, requiring respondents to construct or make explicit aspects of their epistemic cognition that many scholars believe to be largely tacit, complex, and contextual (Chinn & Rinehart, 2016; Sandoval, 2012; Sinatra & Chinn, 2011). Finally, despite attempts to use self-report instruments to examine within-person differences in epistemic cognition across disciplines (e.g., Buehl et al., 2002; Greene et al., 2008), there are many researchers, predominantly those who study epistemic cognition within disciplinary education, who claim these instruments require assumptions that epistemic cognition is traitlike rather than a function of context (Elby & Hammer, 2010). To some degree, this claim about the inherent assumptions regarding the use of self-report instruments is borne out in practice: The vast majority of studies conducted with self-report instruments like the EQ or the Epistemic Belief Inventory involve designs where participants complete self-report instruments in artificial contexts (e.g., completing a survey individually, without being asked to engage in a learning task), as opposed to more authentic contexts that actually invoke epistemic cognition (e.g., arguing about the meaning of experimental data; Sandoval, 2012).

Various forms of semistructured interviews have been used to measure epistemic cognition, beginning with Perry's (1970) seminal work and continuing through many of the studies on developmental models of epistemic cognition (Mason, 2016). For example, such interviews were used extensively over the course of King and Kitchener's (2004) scholarship on their reflective judgment model; they involved presenting participants with ill-structured problems about which "reasonable people reasonably disagree" (p. 5), such as whether news reporting is accurate or chemical additives to food are safe. These interviews were concurrent, meaning that a scenario or example was presented to participants, who then had to respond to interview probes about it, often "without practice or even much time to collect their thoughts" (King & Kitchener, 2002, p. 57). Like the majority of research using self-report instruments, studies involving concurrent semistructured interviews tend to be conducted outside actual learning or practice contexts.

Recently, there has been an increase in the number of studies designed to capture aspects of developmental and dimensional models of epistemic cognition while in the context of thinking about everyday problems or scenarios. For example, there has been an increase in the use of concurrent think-aloud protocols (Ericsson & Simon, 1993) to surface participants' epistemic cognition as they engage in everyday epistemic tasks such as searching the Internet for information (e.g., Barzilai & Zohar, 2012; Ferguson et al., 2012; Greene, Yu, & Copeland, 2014; Hofer, 2004; Mason et al., 2010, 2011). Cognitive interviews have also been used to assess the ways in which people understand the items on survey instruments (Greene, Torney-Purta, Azevedo, & Robertson, 2010), as well as the ways in which their thinking might challenge assumptions of dimensional and developmental models of epistemic cognition (Greene & Yu, 2014; Muis, Duffy, Trevors, Ranellucci, & Foy, 2014). Importantly, however, the majority of this research involving concurrent measures of epistemic cognition has been conducted outside of formal learning (i.e., classrooms) or practice contexts.

On the other hand, many authors from a disciplinary education background have argued that epistemic cognition should be studied in authentic practice contexts, such as when students conduct experiments or construct models (Elby & Hammer, 2010; Kelly, 2016; Sandoval, 2005). Observation and discourse analysis are two predominant nonintrusive assessment methods that allow data collection while students engage in authentic tasks (Kelly, 2016). The data that arise from these techniques are often context-specific and interpreted solely within a single model of epistemic cognition, which makes it difficult to make inferences from them to other developmental or dimensional models (Sandoval, 2005).

Observation and discourse analysis can be done nonintrusively, but some researchers have argued that the addition of retrospective interviews to these designs can lead to more informed understanding of people's epistemic cognition (Chinn et al., 2011; Sandoval, 2005). For example, Pluta, Chinn, and Duncan (2011) assessed epistemic cognition by first asking participants to engage in scientific inquiry and then prompting them to reflect on their epistemic practices and ideals. Data from these retrospective interviews can be triangulated with observation and discourse analysis data

gathered during practice to understand both enacted epistemic practices and the epistemic aims and ideals that students bring to those practices (e.g., Chinn & Rinehart, 2016; Sandoval, 2012). Hammer and Elby have studied their epistemological resources model by triangulating across observational studies of activity (Rosenberg, Hammer, & Phelan, 2006) and cognitive interview studies (diSessa et al., 2002).

There are a number of scholars who examine how people think about knowledge and knowing across and outside of formal learning contexts (Bricker & Bell, 2016). While these researchers may not identify their work as epistemic cognition research, this area of scholarship makes an important, unique contribution to the field. The models of epistemic cognition providing the foundation for this area of scholarship vary, from developmental models (e.g., Weinstock, 2011) to epistemological resources models (e.g., Bang & Medin, 2010). As would be expected given the research focus, these scholars study epistemic cognition in a wide variety of contexts, including courtrooms (Weinstock, 2016), Internet browsing (Strømsø & Bråten, 2010), and museums (Bell, Bricker, Reeve, Zimmerman, & Tzou, 2012). Capturing epistemic cognition in situ often requires open-ended assessment methods and nonintrusive research designs, such as observations and case studies, although some of this research includes interviews either during or after observing epistemic cognition in practice. Research across contexts has revealed how people enact epistemic cognition differently in response to these contexts (Goldman, 2011; Gottlieb & Wineburg, 2012). Greene (2016) argued that within-person differences in enacted epistemic cognition may be due, in part, to people's perceptions of the salient epistemic system, or systems, across contexts (Goldman, 2011). Students who are hesitant to enact epistemic practices in the classroom may do so eagerly in a museum, and their perceptions of the epistemic systems at play may hold the key to understanding these differences in performance across contexts. Scholarship on these contextual differences illustrates how multiple assessment methods are needed to capture not only how epistemic cognition happens but also how features of context are perceived as relevant (Bricker & Bell, 2016).

Our review of research designs and methods in epistemic cognition has revealed a number of trends concerning the assessment methods used, the contexts in which research is conducted, and the kinds of epistemic cognition under study. Continued psychometric problems with self-report methods of assessing epistemic cognition, coupled with the often-decontextualized or inauthentic contexts in which these data are gathered, and their presumptions of domain consistency or traitlike aspects of epistemic cognition, cast doubt on research findings from these studies. The growing acceptance that epistemic cognition is largely tacit, and that people's enactment of epistemic cognition can often be different from their expressed reconstruction of that cognition, has led an increasing number of researchers to pursue concurrent assessment methods (e.g., observations, discourse analysis, think-aloud protocols) that can be implemented in contexts where participants are asked to think about knowledge and knowing (e.g., inquiry activities, searching the Internet for information on controversial issues). Researchers' models of epistemic cognition seem to predict their choice of context, with adherents to dimensional or developmental models often

investigating how people enact epistemic cognition outside of practice settings, and adherents to situative models of epistemic cognition more often using various kinds of inquiry contexts as research settings (Mason, 2016). An important question being studied by researchers of everyday epistemic cognition is the degree to which research in one of these contexts (e.g., classrooms, museums, courtrooms) can connect to, or inform, research in other contexts. Regardless of method, research design context, or underlying model, it is apparent to us that there is a clear need for research involving multiple, mixed methods of gathering data, across a variety of contexts, to understand how epistemic cognition is acquired, enacted, and changed (Chinn et al., 2011). Fault lines across methodology seem to be the most amenable to a “more is more” approach.

RESOLUTIONS LOOKING FORWARD

To close this review, we offer some possibilities for how current fault lines in research on epistemic cognition can be bridged. One possibility is to expand and coordinate across sites of intervention around epistemic cognition. A second is to carefully, and perhaps more openly, imagine possibilities for measuring or describing epistemic cognition in the many places where it occurs. Third, it seems clear that researchers working on questions of epistemic cognition must conscientiously cross disciplinary boundaries to synthesize and extend coherent and reliable findings and models. Addressing each of these possibilities, however, may require integrating conceptual perspectives on epistemic cognition.

Integrating Perspectives on Epistemic Cognition

A basic issue to resolve, from our reading of the many fields that now comprise epistemic cognition research, is the relation between traditional cognitive accounts focused on individual cognitive structures and processes and situative accounts of persons functioning within systems of cultural activity. Greeno (2015) argued for a notion of “integrative pluralism” that accepts that cognition can be seen as occurring, and thus modeled, at multiple levels. He asserted that models focused at different levels of aggregation can each advance understanding of important phenomena. Beyond mere pluralism, however, is the goal of integrating alternative conceptualizations. Greeno suggested that a single, unifying theory that accomplishes such integration is obviously desirable but not currently feasible. He argued instead for case studies that can integrate explanations for some aspects of functioning across cognitive and situative levels. Given the scope of contexts where epistemic cognition is now seen to occur and the various perspectives on how it happens, integration of findings from these myriad contexts should be a major focus of research in the coming century.

Progress toward an integrative pluralism of accounts of epistemic cognition can occur through an expansion of current trends in epistemic cognition research. Obviously, to integrate theoretical accounts across multiple levels of analysis requires an articulation of the levels. Greeno (2015) suggested there are at least three levels relevant to cognition generally: the individual, the individual-in-interaction, and the

activity system(s) in which the interaction occurs. Much of the research comprising the data set for this review focuses on the individual, on characterizing the structure of epistemic beliefs and their function. Some of it attends more closely to individuals-in-interaction from a situative perspective, by attending to how material and social resources available within a given interaction mediate the epistemic cognition engaged by individuals. We think that relatively little epistemic cognition research to date focuses at the level of activity systems, in other words, at the level of communities of people and the practices collectively used to solve epistemic problems. There are studies of episodes of collective activity and its mediation (e.g., Kelly, 2016; Rosenberg et al., 2006). Muis and Duffy's (2013) construct of epistemic climate is an attempt to characterize an aspect of an intact activity system, a classroom, in relation to its influence on the epistemic cognition of actors in that activity system. Greene's (2016) discussion of interacting epistemic systems, and how they affect individuals in those systems, is another way of conceptualizing effects among different levels. We encourage more work at the activity system level as an avenue to increase our understanding of how particular communities of practice might structure epistemic cognition. Kelly (2016) points to sociological studies of science as a helpful analog for such work.

Integrating across levels of analysis further requires that individual researchers, or teams, are at least aware of, if not specifically attentive to, work and models at levels that are not familiar. For example, researchers focused on individual cognitive structures and their assessment should be mindful of how particular research settings may both shape the goals of participants and structure the social and material resources that might mediate epistemic cognition in that setting. Researchers looking at the participation of individuals in collective activity could consider how to trace the individual cognitive consequences from that participation. We see such efforts beginning to happen, and likely to continue, across sites of intervention and measurement of epistemic cognition.

Bridging Schooled and Unschooled Contexts

In the past decades, epistemic cognition researchers have made considerable headway in understanding how more adaptive epistemic cognition can be promoted at different educational levels (Bråten, 2016). In elementary classrooms, intervention work in science (e.g., Metz, 2011; Ryu & Sandoval, 2012), mathematics (e.g., Mason & Scrivani, 2004; Verschaffel et al., 1999), history (Nokes, 2014; VanSledright, 2002), and language arts (Reznitskaya et al., 2012) suggests how engagement in scaffolded disciplinary practices can change students' beliefs and thinking about knowledge and knowing in these disciplines. Moreover, many studies have tried to induce epistemic change in secondary and postsecondary classrooms, focusing on different disciplines (e.g., Bell, Matkins, & Gansneder, 2011; Hynd-Shanahan, Holschuh, & Hubbard, 2004; Muis & Duffy, 2013). Whether any changes brought about by such interventions affect people's beliefs and thinking about knowledge and knowing outside formal educational contexts is currently not known, however (see Bendixen, 2016; Muis, Trevors, & Chevrier, 2016).

One important aspect of epistemic cognition that is highly relevant outside school is how people evaluate conflicting knowledge claims about complex scientific and socioscientific issues (Bromme & Goldman, 2014; Tabak, 2016), for example, when searching for information on the Internet (Strømsø & Kammerer, 2016). In such instances, people's critical-analytic evaluation of sources becomes a key competence, yet this is a competence that many researchers, educators, and policymakers around the world feel is not adequately developed through schooling, not even at the secondary and postsecondary levels (Bråten & Braasch, *in press*). Some recent studies (e.g., Braasch et al., 2013; Kammerer, Amann, & Gerjets, 2015; Mason, Junyent, & Tornatora, 2014; Stadtler, Scharrer, Macedo-Rouet, Rouet, & Bromme, 2016) suggest that people's ability to cope with conflicting knowledge claims about scientific and socioscientific issues can be improved through brief source evaluation interventions. However, next to nothing is known about whether or how epistemic change resulting from participation in disciplinary practices in school can be brought to bear on students' judgments of disciplinary issues in everyday contexts, highlighting the issue of transfer across schooled and unschooled contexts. There is thus a clear need for future research that investigates and seeks to promote connections between epistemic cognition embedded within disciplinary practices in school and thinking about knowledge and knowing outside formal educational contexts. Such research should not be limited to the discipline of science, however.

For example, VanSledright and Maggioni (2016) suggested that citizens will have difficulties participating in genuine democratic discourse and making balanced decisions about the future without developing adaptive epistemic cognition within the discipline of history, involving the weighing of evidentiary support for conflicting accounts and using interpretive decision criteria (e.g., corroboration). To what extent promoting adaptive epistemic cognition through history education will actually contribute to critical, deliberate thinking about controversial social and political issues outside school (e.g., helping people avoid polarization) is an open question for future epistemic cognition researchers to address.

Until recently, epistemic cognition change has been seen as a progression from naïve to sophisticated general views on knowledge. It is now coming to be seen as appropriate adaptiveness to context. A reliance on authoritative sources, for example, is no longer seen as inherently inferior to other ways of knowing, given that much of what we learn comes from the testimony of others (Chinn et al., 2011). Teaching for adaptive epistemic cognition in a way that not only promotes disciplinary learning but also has consequences for how students engage and cope with controversial scientific and social issues outside school requires teachers who know the epistemologies of the disciplines they teach, the applicability of those disciplines to students' everyday lives, and pedagogies that promote connections between epistemic cognition in the discipline and in everyday life. One issue in need of much further research is the potential lack of calibration between teachers' own epistemic cognition and the epistemic aims (e.g., deep understanding) and reliable processes (e.g., reasoned argumentation; Chinn et al., 2014) they try to implement in their classrooms. This is an important issue

because less adaptive epistemic cognition on the part of the teachers may interfere with teaching for adaptive epistemic cognition in students, highlighting the need to support the development of epistemic cognition among teachers participating in epistemic cognition interventions (Brownlee & Schraw, in press). An important goal for epistemic cognition researchers should be to develop empirically based, testable instructional models that can guide interventions for epistemic change (Bråten, 2016). Preferably, such instructional models should also take into consideration the need to help teachers develop their own epistemic cognition and describe how this can be effectively done.

Finally, more intervention work is needed that spans individual and social levels of intervention and coordinates between them. For example, research is needed regarding how to integrate reflection on one's own epistemic beliefs regarding particular disciplinary issues with collective reflection and argumentation as part of social disciplinary practices dealing with those issues. This implies that the effectiveness of epistemic cognition interventions would have to be evaluated quite broadly by attending to changes in disciplinary practices at the interpersonal level, in addition to individual belief change and potential interactions between those loci of intervention (e.g., Ryu & Sandoval, 2012).

Sites of Measurement

Many of the most cited publications we reviewed, from Hofer and Pintrich's (1997) article through more recent reviews and conceptual revisions of epistemic cognition (e.g., Barzilai & Zohar, 2014; Chinn et al., 2011), have lamented the poor reliability and lack of construct validity for scores from Likert-type scale instruments. Numerous studies have shown that the most frequently used self-report measures of epistemic cognition do not meet even minimal psychometric criteria, particularly in terms of configural invariance across samples and contexts (e.g., DeBacker et al., 2008). There appears to be a growing consensus that the ease of administering Likert-type scale assessments of epistemic cognition comes with concerns about the authenticity of whatever it is that those instruments purport to measure (Hofer & Sinatra, 2010; Sinatra, 2016). The field should no longer tolerate publications that utilize instruments with low reliability, poor data-model fit, or even worse, no investigation of construct validity at all. There is nothing innately flawed about quantitative methods of measuring epistemic cognition, but developing, testing, and refining such measures requires effort, time, patience, and persistence. Researchers interested in developing such measures should endeavor to understand and meet epistemic ideals for psychometric validity (e.g., Bandalos & Finney, 2010; Knapp & Mueller, 2010) and should fully report the evidence regarding such efforts in their empirical work. Editors, reviewers, and publishers should be critical of manuscripts that do not meet these ideals.

From our vantage point, epistemic cognition research needs a "right tool for the right job" perspective. People can self-report best about thoughts and attitudes that are explicit in their minds and that require little construction in the moment. Whether self-report instruments should use Likert-type scales or other forms of

response (e.g., semantic differentials; Stahl & Bromme, 2007) is an open question. On the one hand, epistemic cognition phenomena that may be amenable to self-report assessments include epistemic metacognitive knowledge and epistemic aims (Barzilai & Zohar, 2014; Chinn et al., 2011). On the other hand, collecting data about what people do when enacting epistemic cognition requires that measurement occur in contexts where people actually enact epistemic cognition (Kelly, 2016). Analyzing data as people engage in epistemic practices, including observations of the practices used, as well as of the discourse between people (e.g., Herrenkohl & Cornelius, 2013; Ryu & Sandoval, 2012; Samarapungavan et al., 2006), can be a powerful way to understand not only people's epistemic cognition but also their perceptions of larger epistemic systems such as science or history (Goldman, 2011; Greene, 2016), as well as how those systems enculturate people to normative practices and knowledge (Kelly, 2014).

One benefit to observations is that they can be used to collect data without interfering in people's epistemic cognition. However, the bridge from observation data to research findings is inference, which often can be difficult to justify. The challenges of deriving and justifying inferences increase as researchers move from studying what people do (e.g., epistemic practices), to studying what they think but often do not verbalize (e.g., epistemic vices). Even more difficult are inferences about what people have learned but rarely reflect on (e.g., epistemic metacognitive skills, epistemic ideals), and what they feel but rarely think about (e.g., epistemic metacognitive experiences; Barzilai & Zohar, 2014; Chinn & Rinehart, 2016). Epistemic cognition phenomena that are rarely consciously invoked may be best measured by prompting individuals to surface and reflect on the otherwise nonconscious aspects of their epistemic cognition (e.g., "How did you feel when you realized that your views on evolution were being challenged?"). Regardless of whether researchers intervene into participants' epistemic cognition or not, these data can and should be usefully triangulated with data from semistructured interviews (e.g., Feucht & Bendixen, 2010), cognitive interviewing (e.g., Barzilai & Weinstock, 2015; Greene, Torney-Purta, Azevedo, et al., 2010; Muis et al., 2014), or cued reflection (e.g., Berland & Crucet, 2016; Ferguson, Bråten, Strømsø, & Anmarkrud, 2013), as well as data regarding the context in which the participants engaged in epistemic cognition (e.g., Kelly & Crawford, 1997).

Of course, the more unusual the context, the more the findings are encapsulated. Completing a survey before a class begins is not something that students do very often; therefore, it is not surprising that data from this context have shown only modest-to-small relationships with data drawn from more familiar contexts, such as grades in a class. What is needed is more description of the contexts in which epistemic cognition is measured, and more research that systematically varies context in investigations of the same epistemic cognition phenomena. Such studies would reveal how perceptions of contextual factors, such as epistemic climate (Feucht, 2010; Muis & Duffy, 2013), influence enacted epistemic cognition. Likewise, drawing from psychometrics research, there is a need for studies that assess the same epistemic

cognition phenomena across contexts designed to elicit typical performance (e.g., casual browsing of the Internet for a question of curiosity) and maximal performance (e.g., heavily scaffolded or supported investigations of questions with serious implications, such as reviewing literature for a dissertation).

In sum, the field's growing recognition of the need for more attention to context and tools of measurement is not a sign of methodological or philosophical splintering, nor does it warrant calls for nihilism or methodological relativism. Rather, it represents the field's better understanding of the problem space and of the complexity of the phenomena under study. To borrow from Perry (1970), we suspect that in the coming years, the field will make commitments within methodological relativism, where different combinations of epistemic cognition phenomena and contexts will be addressed with different measurement tools, which will be seen as more or less useful depending on the circumstances.

Transdisciplinary Possibilities

Academic disciplines tend to fragment themselves in order to focus on specific questions and problems, and psychology and education are no different in this regard. Furthermore, education is a huge field informed by multiple disciplinary perspectives besides psychology, including anthropology, sociology, history, and more. These perspectives often clash around theories, methodologies, and ideologies. Nevertheless, it is clear that scholars working on issues of epistemic cognition recognize the need to bridge disciplinary boundaries, and this trend will, and should, continue. As we have shown in this review, perspectives from disciplinary education research have reframed questions of the general versus specific character of epistemic cognition. This has come from now well-established lines of communication between educational psychologists, disciplinary education researchers, and learning scientists working within those disciplines. The next step to take in this dialogue is to try to synthesize empirical results across subfields to scrutinize available models of epistemic cognition. A recent example of this effort includes a broad synthesis of research across the elementary grades in math, science, and social studies as those areas of teaching might bear on the theory of integrated domains in epistemology model of epistemic cognition (Bendixen, 2016), an effort notable for its attention to substantive differences among disciplines and their significance to modeling epistemic cognition broadly. More syntheses like these ought to be conducted in the coming century.

Extending this sort of synthesis requires several things. Fundamentally, it requires researchers to be knowledgeable about the many fields of research that bear on specific questions of epistemic cognition. This demands both training across disciplines that would seek to develop integrative specialists around particular aspects of epistemic cognition, and research collaborations that cross specializations and can produce the sorts of case studies that Greeno (2015) suggested. It may also be the case that epistemic cognition research needs particular lines of communication to become more open. For example, the links between relevant text comprehension research (e.g., Bråten, Strømsø, et al., 2009) and disciplinary education research are relatively weak,

and the same is likely to be true for other subfields of epistemic cognition research. Finally, it is critical that integrative work explicitly tests predictions of various models of epistemic cognition against one another. Epistemic cognition research over the past century has seen only a profusion of possible models, with no real pruning of the contenders. While a single theory of epistemic cognition and its development seems infeasible now, the work of comparing the value of alternative models should be done.

CONCLUSION

Over the past century, epistemic cognition has moved from a latent concern to a buzzing, blooming field of research. Our review has traced the many historical lines of scholarship that have produced the buzz, and the emerging confluence of these lines into a broad, perhaps turbulent, stream of research. We are optimistic that recent trends of dialogue between the fields interested in epistemic cognition will expand over the next century to generate more integrative approaches to the study of epistemic cognition and better accounts of its development. Such trends must take into account multiple levels of analysis, the many forms and causes of development, and how the context in which research occurs can influence the types of epistemic cognition produced. Only then will the field have what it needs to effectively promote the kinds of thinking needed to successfully traverse the many challenges of the 21st century and beyond.

NOTE

¹We also do not attempt any summary of history, philosophy, or sociology of science, or other disciplines, even though those fields have significantly influenced contemporary epistemology and the ways in which disciplinary education research frames and studies epistemological issues. (Interested readers can consult Chinn et al., 2011; Depaepe et al., 2016; Kelly, 2016; VanSledright & Maggioni, 2016.)

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